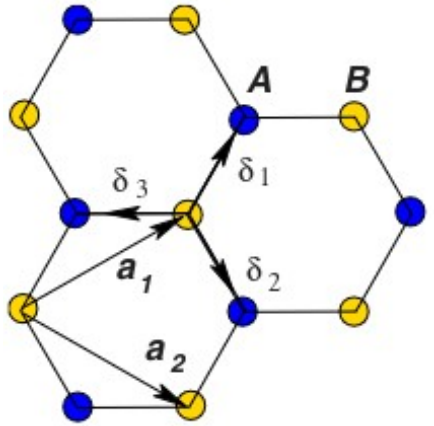
Dresden University of Technology

Eric Kürschner, Matthias Vojta (Advisor)

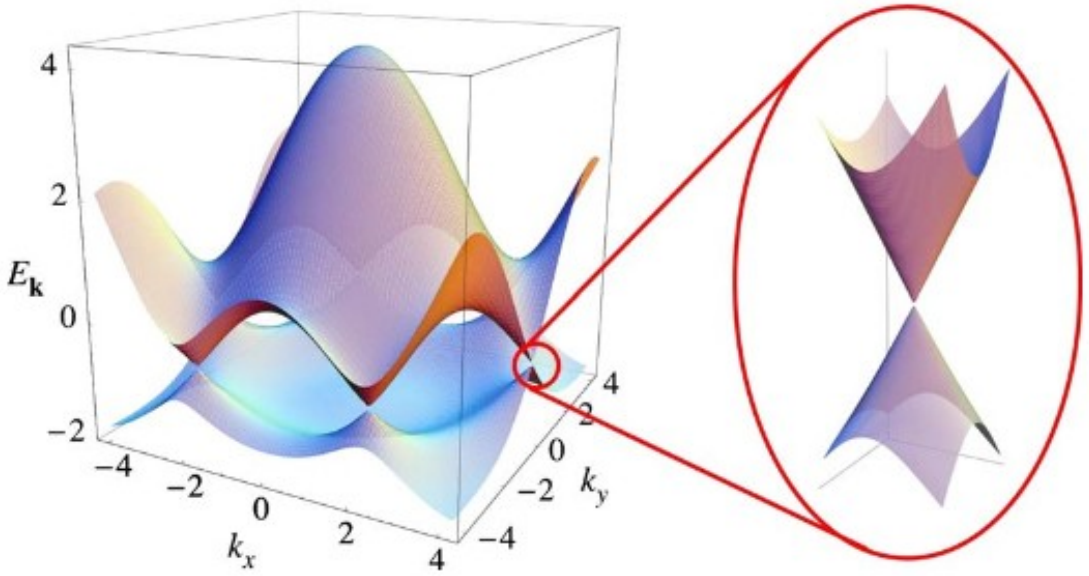
Strain Induced Pseudomagnet Fields in Quantum Spin Liquids

Tight-Binding Description of Graphene

$$H = -\sum_{\langle i,j \rangle} t_{ij} a_i^\dagger b_j + \text{h.c.}$$



ugly empty space



1-dimensional Fermi-surface at K-points

linearized equation of motion near K-points:

$$v_F \vec{\sigma} \cdot \vec{p} \Psi_{\vec{K}} = E \Psi_{\vec{K}}, \quad \Psi_{\vec{K}} = \begin{pmatrix} a_{\vec{K}} \\ b_{\vec{K}} \end{pmatrix}$$

⇒ **massless Dirac-equation**

(Castro Neto et al., 2009)

Straining Tight Binding Graphene: Pseudomagnetic Field

Straining lattice alters nearest-neighbour-bonds

$$\vec{\delta}_{ij} \to (\underline{I} + \underline{u})\vec{\delta}_{ij}$$

$\underline{u}$ … strain tensor

Assuming

$$t_{ij} = t \exp\left[-\beta\left(\left|\vec{\delta}_{ij}\right|/a - 1\right)\right]$$

β … magneto-elasitc coupling

$$\Rightarrow t_{ij} \to \left(1 - \beta\left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}\right)\right) t_{ij}$$

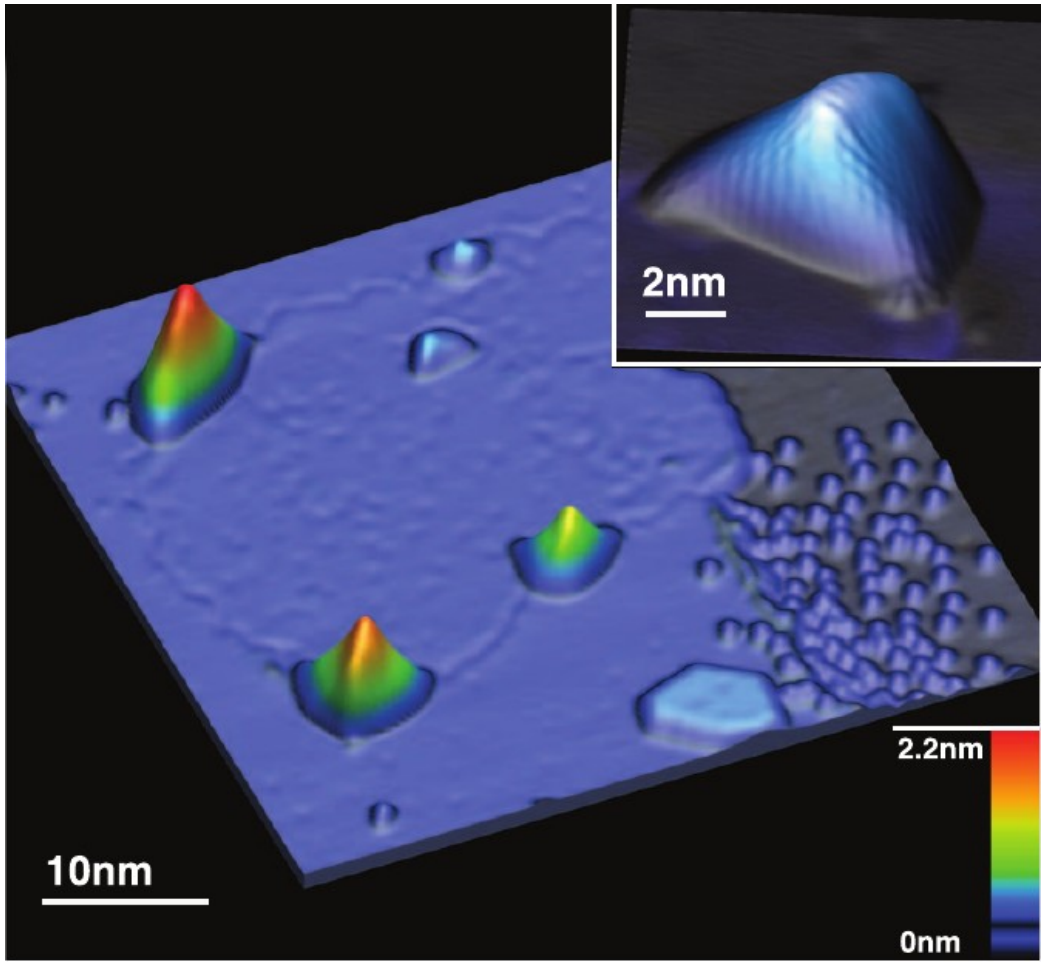
Modified equation of motion:

$$\mathrm{v}_F \vec{\sigma} \cdot \left(\vec{p} + \vec{A}\right) \Psi_{\vec{K}} = E \Psi_{\vec{K}}, \qquad \vec{A} = \frac{\beta}{2}\begin{pmatrix} u_{xx} + u_{yy} \\ 2u_{xy} \end{pmatrix}$$

⇒ **strain-induced pseudomagnetic field** $\vec{B} = \vec{\nabla} \times \vec{A}$

Giant Magnetic Field in Graphene Nanobubbles

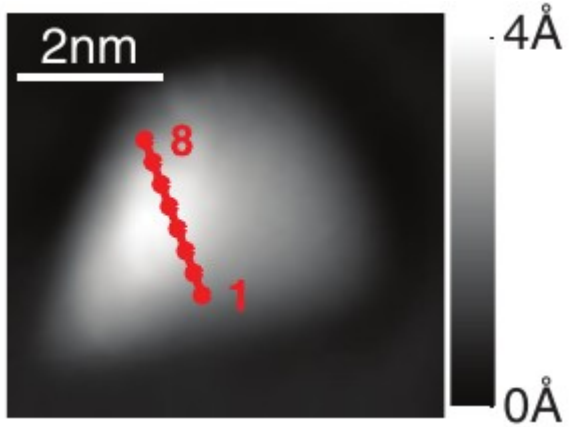
- Graphene Nanobubbles on Pt surface

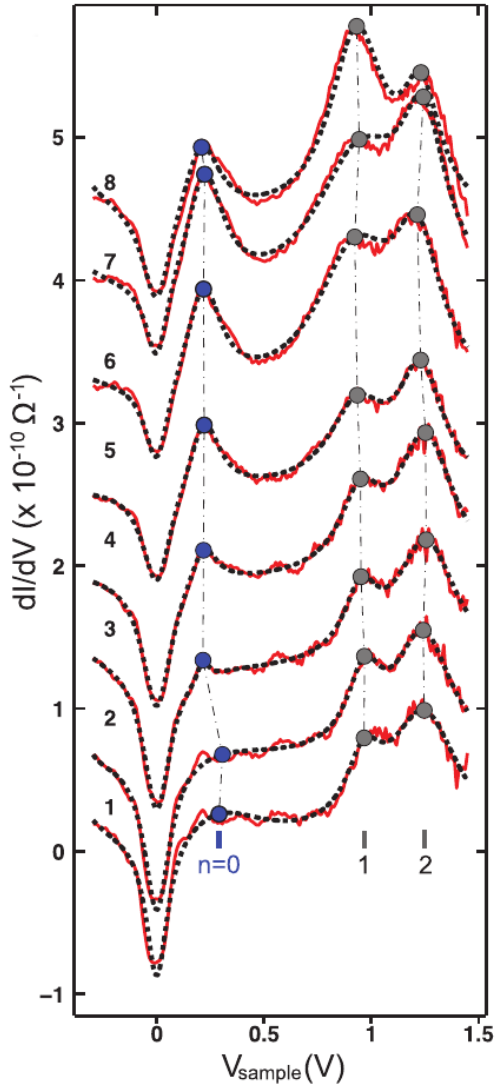


(Levy et al., 2010)

Giant Magnetic Field in Graphene Nanobubbles

- Graphene Nanobubbles on Pt surface
- Scanning Tunnel Spectroscopy probes Local Density of States
  ➔ Landau Level Structure





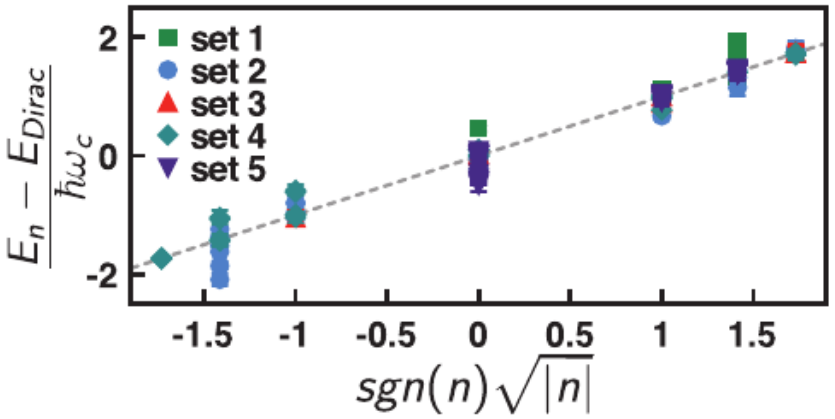
(Levy et al., 2010)

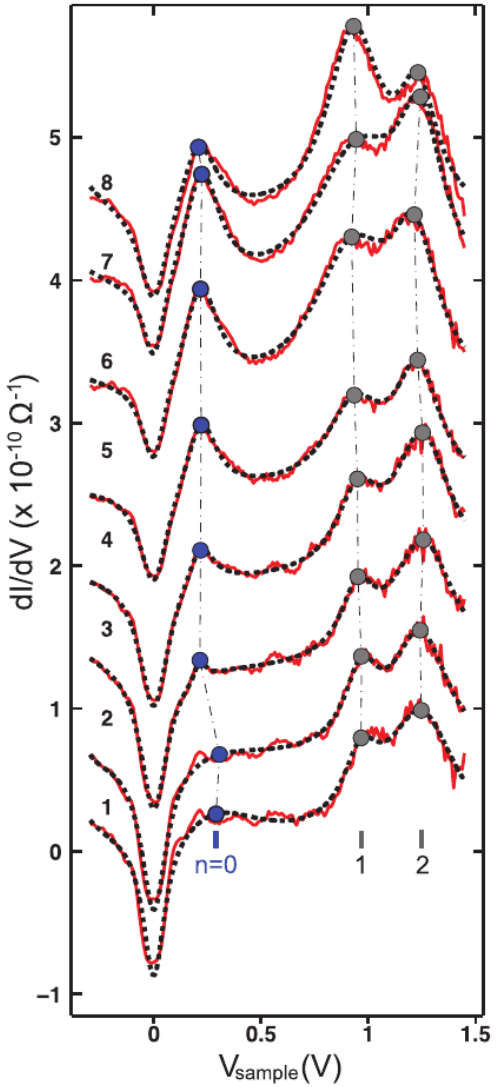
Giant Magnetic Field in Graphene Nanobubbles

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- Fitting Landau Levels with

$$E_n - E_{\mathrm{Dirac}} \propto \mathrm{sgn}(n)\sqrt{|n|B}$$

reveals pseudo-magnetic field strength





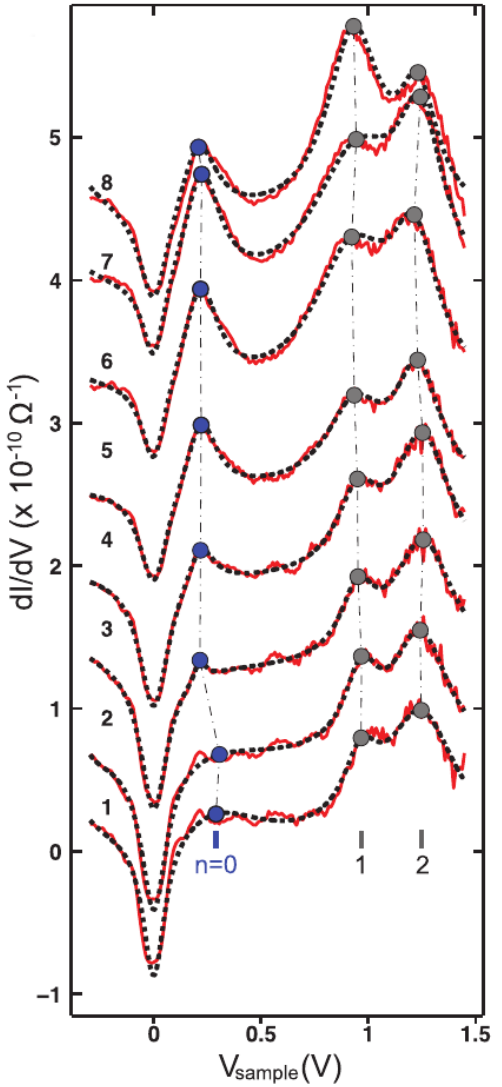
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- Magnetic Field in center of single nanobubbles estimated as 300...400 T
  - ➔ Surpasses any non-destructive magnetic field (record: ~ 100 T, Los Alamos 2012)



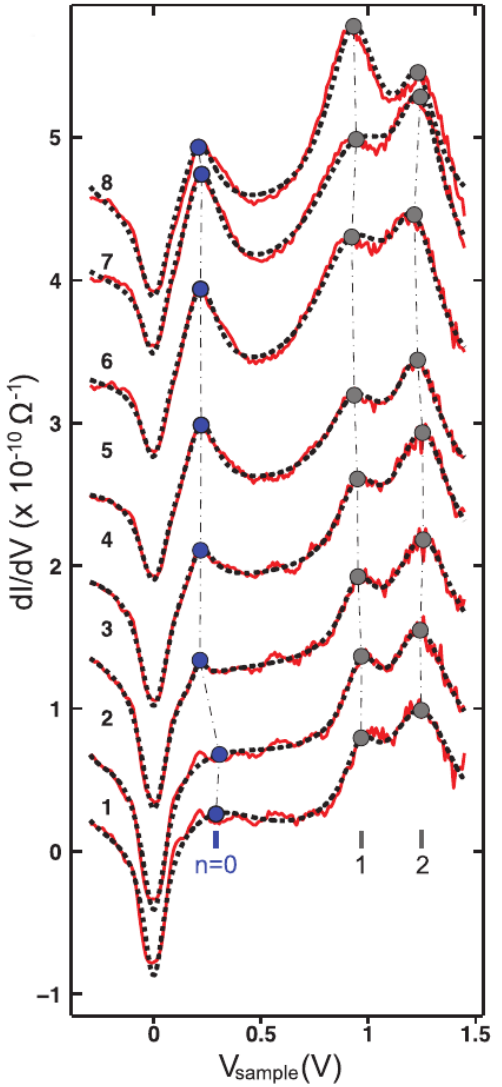
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Giant Magnetic Field in Graphene Nanobubbles

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- Magnetic Field in center of single nanobubbles estimated as 300...400 T
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- Explanation: mechanical deformation of lattice mimics magnetic field

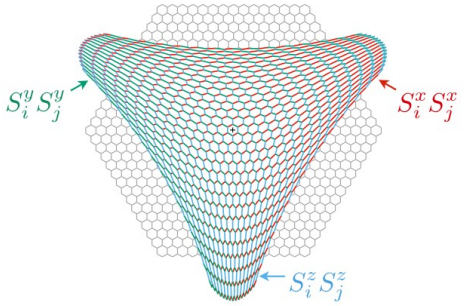


(Levy et al., 2010)

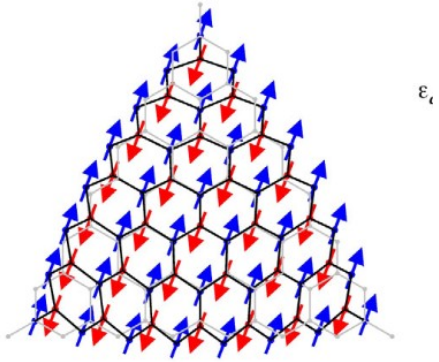
Strain-Enginerring

- novel route for engineering electronic properties
- platform for studying effects in large magnetic fields
- strain-induced pseudomagnetic fields proposed for many other quasi-particles, including

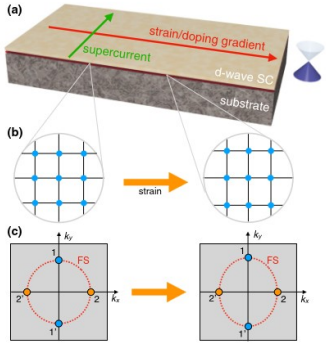
fractional majorana excitations in Kitaev model



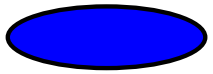
Magnons

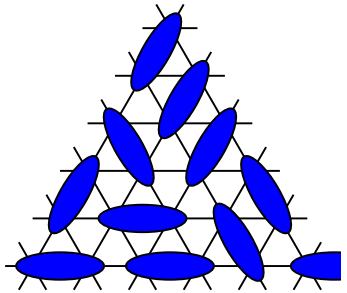


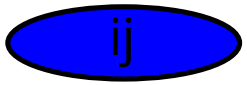
Bogoliubov quasi-particles in super conductors



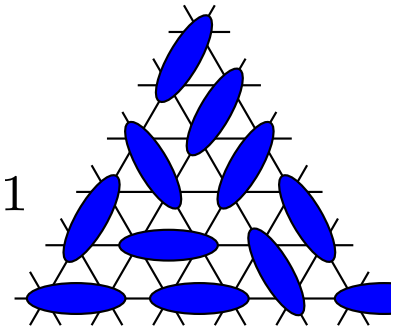
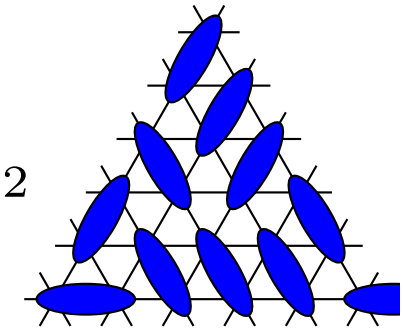
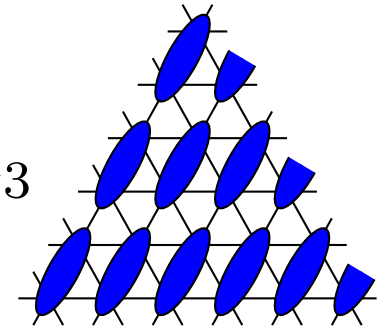
Construction of Resonating Valence Bond (RVB) States

- cover lattice with spin singlets  $= \frac{1}{\sqrt{2}} \left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \right)$



$\hat{=} \prod_{\substack{\text{pairings} \\ (i,j)}}$ 

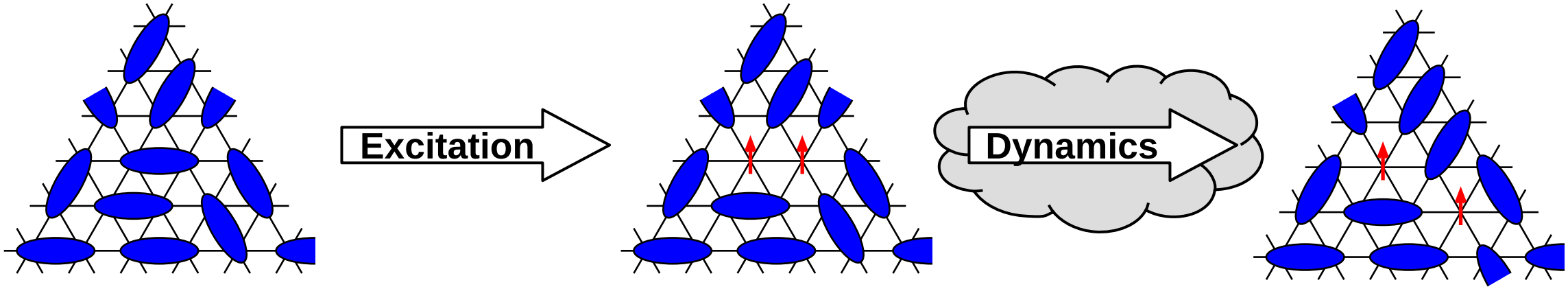
- RVB state is superposition of all possible coverings

$$|\mathrm{RVB}\rangle = a_1$$  $+ a_2$  $+ a_3$  $\ldots$

- certain RVB states do not break lattice symmetry + feature no long range order

Excitations on Resonating Valence Bond (RVB) States

- Excitation: break up Singlet into two un-entangled Spins (called **Spinons**)



excitations move **indepently** of each other

- each Spinon carries fraction of S = 1 $\Rightarrow$ **Fractionalization**

Construction of Parton Mean-Field Theory

Goal: derive effective model of spinons

$$H = \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j$$

- represent Spins as fermionic **partons** (= internal degrees of freedom)

$$\vec{S}_i = \frac{1}{2} \sum_{\alpha\beta} f^\dagger_{i\alpha} \sigma_{\alpha\beta} f_{i\beta}$$

- decouple interacting Hamiltonian by introducing **mean fields** $\chi_{ij} = \frac{1}{2} \sum_\alpha \langle f^\dagger_{i\alpha} f_{j\alpha} \rangle$

$$\Rightarrow H = \sum_{\langle i,j \rangle, \alpha} J^{(\mathrm{MF})}_{ij} f^\dagger_{i\alpha} f_{j\alpha}, \qquad J^{(\mathrm{MF})}_{ij} = -J\chi^*_{ij}$$

Parton Gauge structure

- Parton representation artificially enlarges local hilbert space $\vec{S}_i = \frac{1}{2} \sum_{\alpha\beta} f^\dagger_{i\alpha} \sigma_{\alpha\beta} f_{i\beta}$

$$\{ |\uparrow\rangle , |\downarrow\rangle \} \mapsto \{|00\rangle , |01\rangle , |10\rangle , |11\rangle\}$$

- constrain model to physical subspace (given by $(n_\uparrow + n_\downarrow - 1)\,|\psi\rangle = 0$)

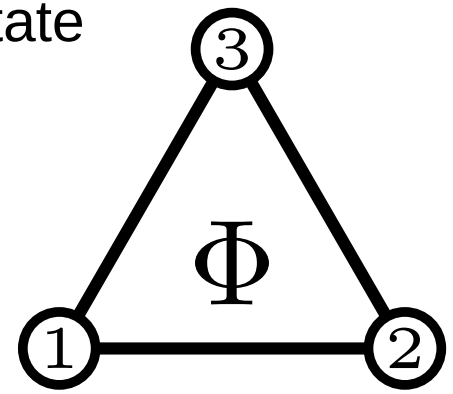
  → leaves behind **gauge redundancy**

$$f_j \mapsto e^{i\theta_j} f_j$$

- Consequence: many (effective) microscopic models describe same QSL state

  → characterize via **gauge invariant plaquette fluxes**

$$\Phi = \arg(\chi_{12}\chi_{23}\chi_{31})$$



Quantum Spin Liquids: broader picture

- System of highly correlated spins that does not order, even at T=0.
- Other prominent features:
  - → no spontaneously broken symmetries (e.g. spin rotation, translation)
  - → fractional excitations (e.g. spinons)
  - → large entanglement may support topological order

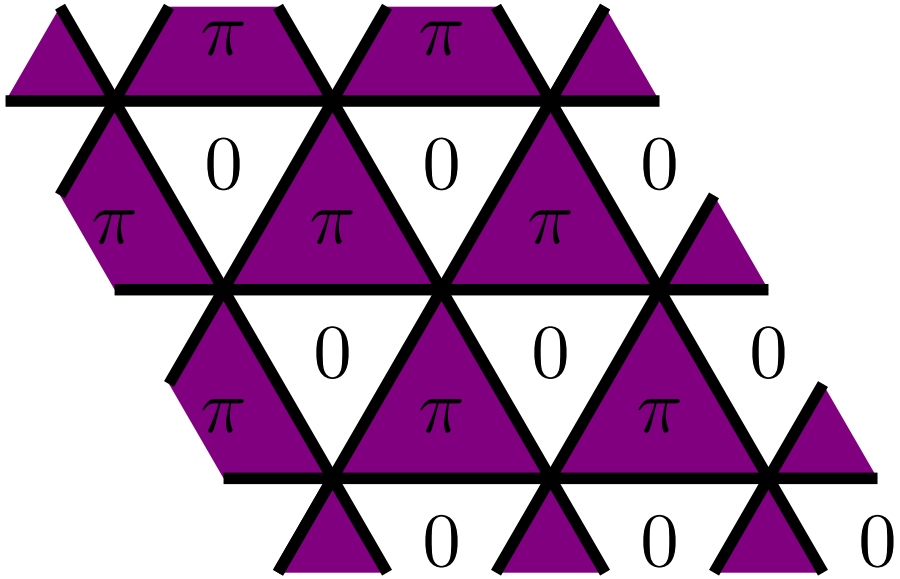
What do I do with this space?

Mean Field Theory on the Triangular Lattice

$$H = J \sum_{\langle i,j \rangle} \left(\vec{S}_i \cdot \vec{S}_j + \kappa \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \right)$$

κ-term to protect translational symmetries

missing: picture of dipsersion to shwocase dirac cones



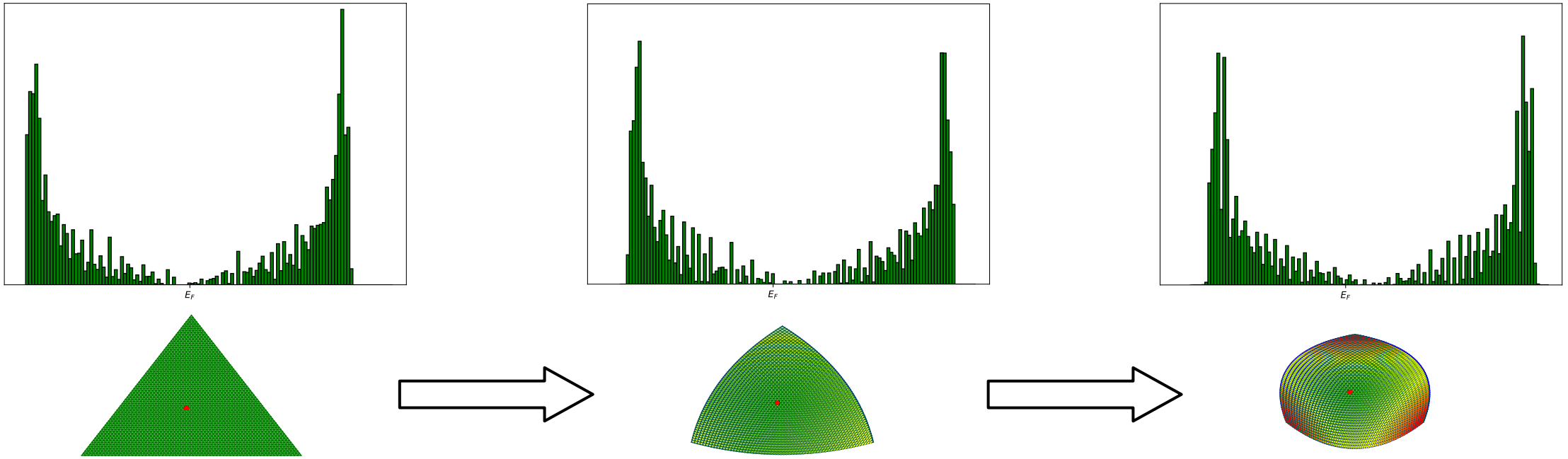
- target state: π-flux pattern; features 2 Dirac cones in 1. BZ
- mean field theory opens gap by breaking TRS :(

  → working solution: force π-flux state by reducing mean field parameter space

Straining the Triangular Lattice: Numerics

- triaxial displacement pattern $\vec{U} = C \begin{pmatrix} 2xy \\ x^2 + y^2 \end{pmatrix}$
- Local Density of States unveils effects of strain

C = 0 | C = 0.35 | C = 0.65



- **no Landau Levels**

Straining the Triangular Lattice: Continuum Theory

Starting point: tight binding model compatible with flux pattern

$$H = \sum_{\langle i,j\rangle,\sigma} t_{ij}^{AA} a_{i\sigma}^\dagger a_{j\sigma} + t_{ij}^{BB} b_{i\sigma}^\dagger b_{j\sigma} + \left(t_{ij}^{AB} a_{i\sigma}^\dagger b_{j\sigma} + \text{h.c.} \right)$$

1. large wavelength expansion around Fermi points reveals continuum theory

Fermi points

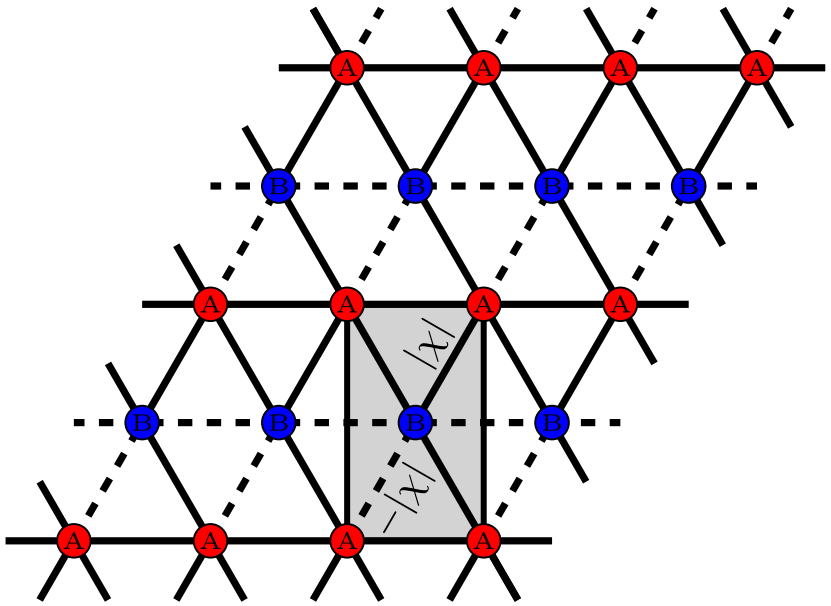
$$a_{i\sigma} \approx e^{i\vec{D}\vec{R}_i} A_{D\sigma}(\vec{R}_i) + e^{i\vec{D}'\vec{R}_i} A_{D'\sigma}(\vec{R}_i)$$

slowly varying fermionic fields

2. expand hopping amplitude in strain

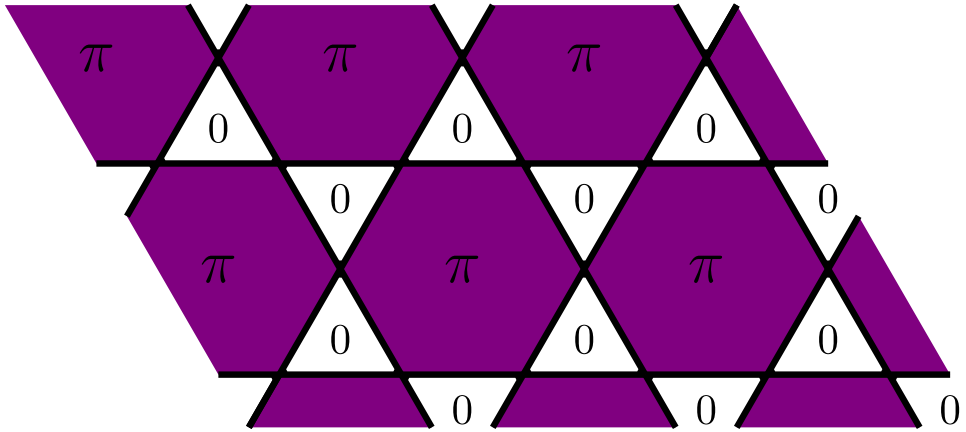
$$t_{ij} \mapsto t_{ij} \left(1 - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \right)$$

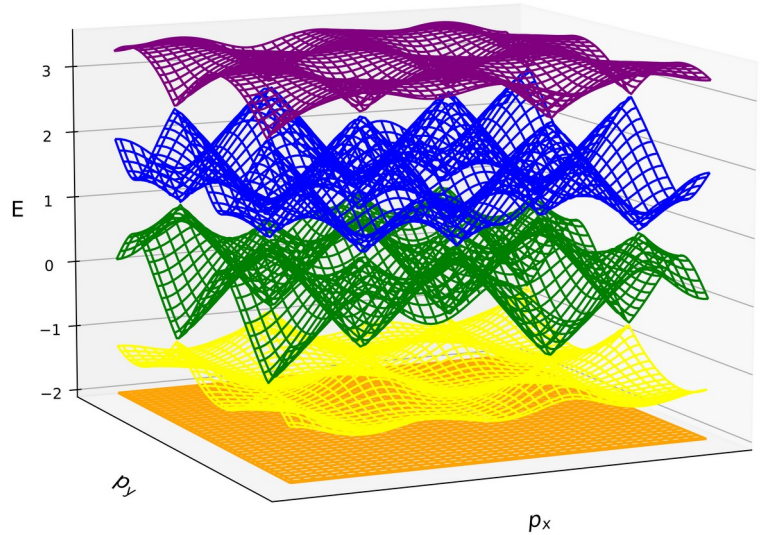
Result: no strain-induced pseudomagnetic field



π-Flux State on the Kagome Lattice

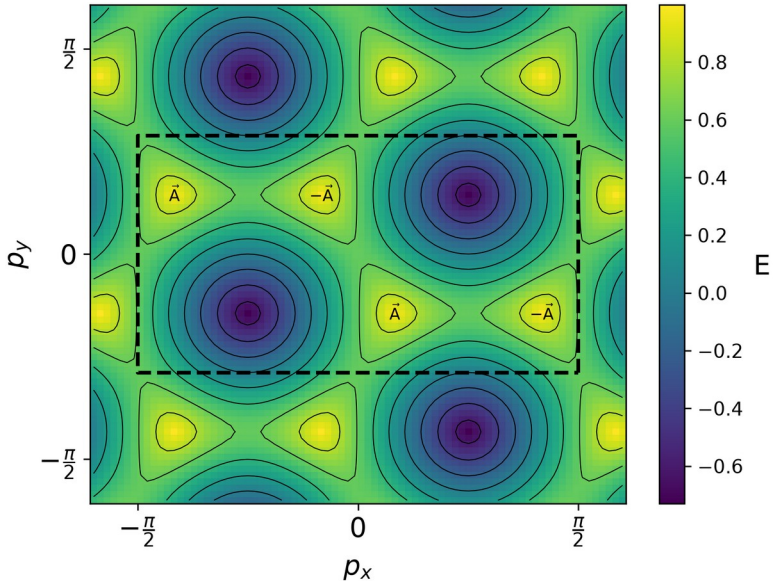
- six bands feature dirac-cones at three different fillings (3/6, 4/6, 5/6)





- pseudomagnetic field only at 4/6-filling

$$\vec{A} = \frac{\beta}{2\sqrt{3}} \begin{pmatrix} u_{yy} - u_{xx} \\ u_{xy} + u_{yx} \end{pmatrix}$$



Summary

- mean field theory on triangular lattice
  → ground state breaks TRS (topological order)
  → π-flux state shows no pseudomagnetic field, verified by continuum theory
- continuum theory on kagome lattice
  → strain induced pseudomagnetic field at 4/6-filling

Outlook

- Probe Landau Level structure using response quantities.
- Investigate interplay of mean field and strain. (resonance effects?)

Advanced questions

- Why do only some states show pseudomagnetic fields?